STAT (2022)

Short Question

what is Yate correction for continuty?

Yates correction for continuity is a statistical adjustment used when approximating discrete data with a continuous distribution.

In a simple terms, it's a fix to make calculations more accurate. when calculating chi-squared tests, You Subtract 0.5 from the difference between observed and expected value before squaring.

$$\frac{(|Observed - expected| - 0.5)^2}{expected}$$

This help get more accurate results, especially with small Sample.

Q(ii) B

write down contigency table for two attributes A & B what are the criteria for their independ

	B present	B absent	Total
A present	a	b	a+b
A Absent	c	d	c+d
Total	a+c	b+d	a+b+c+d

For "A" and "B" independent, the following condition should hold:-

$$P(A \cap B) = P(A) \times P(B)$$

In term of contigency table, this translate to:

$$\frac{a}{a+b+c+d} = \frac{(a+c)(a+b)}{(a+b+c+d)^2}$$

OR equivelently:-

$$a \times (a+b+c+d) = (a+c)(a+b)$$

This can furthor simpliefie to :-

$ad = bc$

so, if $ad = bc$, attributes A and B are likely independent.

Q iii

what is meant by fisher least significance difference test and how is it used in interpreting the result of an experiment?

Fisher least significant Difference test is a statistical method used to compare multiple group after an Analysis of varience.

In simple term,

ANOVA check if there's a significant difference between group mean.

if ANOVA show a difference, Fisher's LSD test identify which specific group differ from each other.

usefullness :-

Identifying speci

group difference.
Help interpret ANOVA resu...

For example:-
Comparing yield of different crop varities. ANOVA sh... a difference, and LSD test identi... which varieties have significantly diffrent yield.

Q(iv)

Five Fertelizer A, B, C, D and E were tested in 5x5 Latin square design. following information is given:

Sum of square	df	Mean square	F...
		49.15	
46.67			
		3.51	
		11.8892	

...te the analysis by filling

In term ...
translate ...

$\frac{a}{a+b+c+d}$ = ...

OR equiv...

a x (a+b...

5x5

in the blanks.

SOV	Sum of Square	DF	Mean Square
Fertilize	196.6	5-1=4	49.15
row	46.67	5-1=4	11.6675
column	3.51x4=14.04	5-1=4	3.51
Error	28.03	4x4=16	28.03/16 = 1.75
Total	11.8892 x 24 ~~24~~ 285.3408	25-1=24	11.8892

Now,

$$\text{Error SS} = TSS - [RSS + CSS + FSS]$$

$$= 285.3408 - [14.04 + 46.67 + 196.6]$$

$$= 28.03$$

$$\text{Sum of Square} = \text{Mean square} \times df$$

$$\text{Mean square} = \frac{\text{Sum of Square}}{df}$$

- [illegible]
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what are the basic principle of experiment design?

The Basic principle of experiments designs are randomization replication and local control. These principl make a valid test of significant possible. Each of them is describe

- Randomization
- Replication

what is meant Analysis of varience (ANOVA)?

The analysis of varience is a technique that partition the total variation - a term distin from varience and measured by the sum of square of deviations from the mean - into its components parts, each of which is assosiated with a different source of variation. These components parts of varience them analyzed in such a manner that certain hypothesis can be tested.

— (Q(Vii)3) —

what is meant by partial correlation?

Partial correlation measure the relationship between two variable while controlling for the effect of one or more additional variable

In simpler terms:

You want to know the relationship between two variable (X and Y).

Partial correlation help you understand the relationship between "X" and "Y" while keeping "Z" constant.

Example :- Relationship between ice cream sales (x) and number of sunburns (y), controlling for temperature (z).

Q viii

Calculate the multiple correlation $R_{2.13}$;

$b_{23} = 0.67$

$b_{12} = -0.6$ $b_{24} = -0.40$

$b_{13} = 0.27$

$b_{31} = 0.60$ $\frac{r_{21} + r_{23} - 2r_{21} r_{23} r_{13}}{1 - r_{13}}$

$b_{32} = 0.38$

$$R_{2.13} = \sqrt{\frac{b_{21}^2 + b_{23}^2 - 2b_{21} \cdot b_{23} \cdot b_{13}}{1 - b_{13}^2}}$$

$$R_{2.13} = \sqrt{\frac{(-0.40)^2 + (0.67)^2 - 2(-0.40)(0.67)}{1 - (0.27)^2}}$$

$$= \sqrt{\frac{0.16 + 0.4489 - (-0.8)(0.1809)}{1 - 0.0729}}$$

$$\sqrt{\frac{0.16 + 0.4489 + 0.14472}{0.9271}}$$

$$= \sqrt{\frac{0.75362}{0.9271}} = \sqrt{0.8128}$$

$$\boxed{R_{2.13} = 0.90} \cong 0.625$$

(IX)

Describe multiple linear regression with two regressers.

Multiple linear Regression (MLR) with two regressers is a statistical model that predicts a continuous outcome variable (Y) based on two predictor variable (X_1 and X_2).

The general equation is;

$$Y = B_0 + B_1X_1 + B_2X_2 + \varepsilon$$

where;

Y :- outcome variable

X_1 and X_2 :- predictor variable

B_0 :- Intercept or constant term

B_1 and B_2 :- Regression coefficient for X_1 and X_2

ε :- Error term.

Example:- predicting house price (Y) based on area (X_1) and numbers of bedrooms (X_2).

Ex (X)

In a trivariate distribution it is found that:.

$s_1 = 1.0 , s_2 = 0.8 , s_3 = 9.0$

$r_{12} = 0.6 , r_{23} = 0.65 , r_{31} = 0.7$

Find

$$b_{23.1} = \frac{r_{23} - r_{21} r_{31}}{(1-r_{21})(1-r_{31})}$$

$$b_{23.1} = \frac{r_{23} - r_{12} \times r_{13}}{\cancel{(1-(r_{12})^2)(1-(r_{13})^2)} \quad 1 - r_{12}^2}$$

$$b_{23.1} = \frac{0.65 - (0.6 \times 0.7)}{1 - (0.6)^2}$$

$$b_{23.1} = \frac{0.65 - 0.42}{1 - 0.36} = \frac{0.23}{0.64}$$

$$\boxed{b_{23.1} = 0.359}$$

$s_1^2 = \sqrt{1.0} \Rightarrow s_1 = 1.0$

$s_2^2 = \sqrt{0.8} \Rightarrow s_2 = 0.8944$

$s_3^2 = \sqrt{9.0} \Rightarrow s_3 = 3.0$